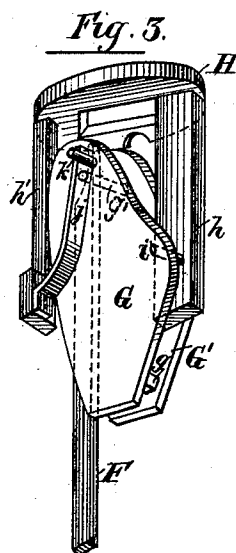
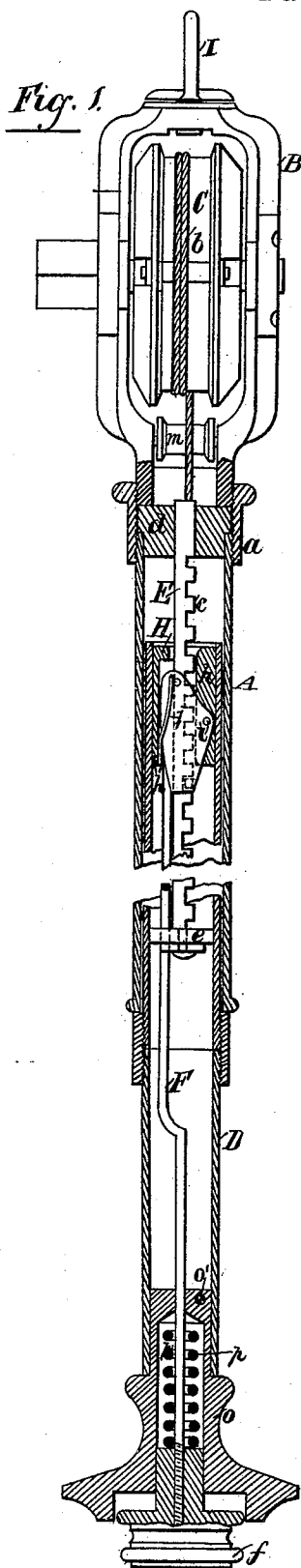
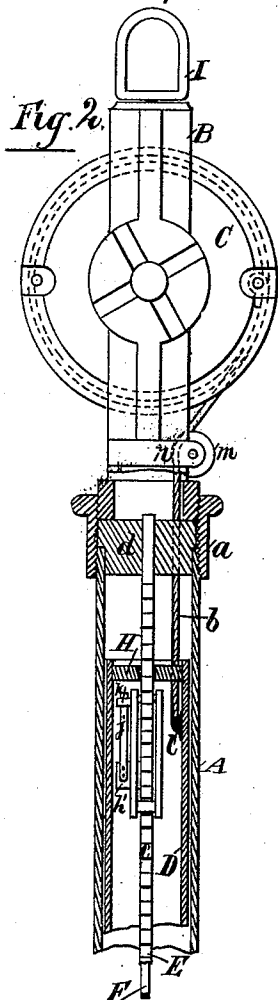


(No Model.)

J. T. BRUEN.
EXTENSION CHANDELIER.

No. 276,130.

Patented Apr. 17, 1883.



Witnesses:

George H. Bott
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Inventor:

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By his Attorney
Ernest Wood.

UNITED STATES PATENT OFFICE.

JOHN T. BRUEN, OF BROOKLYN, ASSIGNOR TO WILLIAM F. WARE, OF NEW YORK, N. Y.

EXTENSION-CHANDELIER.

SPECIFICATION forming part of Letters Patent No. 276,130, dated April 17, 1883.

Application filed March 9, 1883. (No model.)

To all whom it may concern:

Be it known that I, JOHN T. BRUEN, a citizen of the United States, residing at Brooklyn, in the county of Kings and State of New York, have invented certain new and useful Improvements in Extension-Chandeliers, of which the following is a full, clear, and exact description.

The object of this invention is to produce an extension-chandelier or drop-light which can be cheaply made, and will be strong and durable, simple in its construction and operation, and adapted to be easily raised or lowered to bring the illuminating devices carried by it within convenient distances for use.

My invention consists in the combination of a stationary or non-extensible tube, a stationary rack or indented bar, an extensible tube, and a vertical rod offset at its lower end and wedge-shaped at its upper end, whereby the lower end of the rod is adapted to enter and be fixed in a central opening in the shank of a movable button, and its upper end bears against and is adapted, when said button is pressed upwardly, to actuate a pivoted catch or locking device to release said catch from engagement with the rack or indented bar and permit the movement of the extensible tube, and to cause said catch to interlock with the rack or indented bar under an impulse from a spring when the pressure on the button is released; and in the details of construction, all as hereinafter fully set forth.

In the accompanying drawings, in the several figures of which similar letters of reference designate like parts, Figure 1 is a central vertical section of the tubular portion of a chandelier embodying my improvement. Fig. 2 is a side view and vertical section of the upper part of the tubular portion; and Fig. 3 is a perspective view, in detail, of the catch or locking device.

A designates a stationary tube, screwed into a pipe or nipple, *a*, fixed to a yoke, B, carrying the spring-drum C, on which a cord, chain, or strap, *b*, is wound.

D designates an extensible tube, fitted within the tube A, and adapted to slide up and down within the tube A, whereby it may be adjusted to bring the illuminating devices,

which may be suspended therefrom by suitable means, at the desired position for use.

E designates a rack or indented bar fixed to and depending from a collar, *d*, which forms the rack-head held between the end of yoke and top of stationary tube. This rack E is provided with a series of teeth or cogs, *e*, in one of its edges, and is provided at its lower end with a buffer, *e*, which acts as a guide to the rack, and also serves to terminate the extension of the tube D, as will hereinafter appear.

F designates a vertical push-rod, offset near its lower end, whereby it may be screwed into a central opening in the shank of a button or push-piece, *f*, as clearly shown in Fig. 1. The upper end of this rod F is wedge-shaped, as shown by dotted lines in Fig. 3, to actuate a pivoted catch or locking device whereby the movement of the tube D is controlled. The rod F is inserted through an opening in the buffer *e*, and extends upwardly along the smooth edge of the rack E, the buffer serving also as a guide to the rod. This catch or locking device consists of two cam-shaped flat sides, G G', connected by two cross-pieces, *g* *g'*, of a length sufficient to hold the parts G G' far enough apart to permit the free passage of the rack E between them. The cross-piece *g* is square cut and of a diameter sufficiently small to permit said piece *g* to enter the indentations in the rack E between the teeth *e*, and the cross-piece *g'* is rounded off to facilitate the disengagement of the cross-piece *g* with the rack E when the wedge-shaped end of this rod F is brought to bear against said piece *g'*.

H designates an annular ring fitted snugly into the tube D, and provided with two prongs or ribs, *h* *h'*. The prong *h* is longitudinally secured to the tube D, and is straddled by the parts G G' of the catch, which are pivoted thereto, as shown at *i*, Fig. 3, and is of a slightly greater thickness than the rack, whereby it serves as a guide for the same and prevents any lateral play of the rack. The prong *h'* is offset to permit the free play of the catch or locking device, and has attached to it a spring, *j*, which bears against a stud, K, on the part G. The suspending-cord *b* passes through an opening in the annular ring H,

and is fixed to the upper end of the tube D, as shown at *l*, Fig. 2, the other end being fixed to the spring-drum C, and the cord passing over an idler or loose wheel, *m*, grooved to receive it, and hung on a pin fixed to two projections, *n*, on the yoke B, whereby it is guided as it is wound on or unwound from the drum C.

I designate a swivel-buckle attached to the yoke B, by which the chandelier may be suspended from a hook or like device. *o* designates a knob snugly fitted into the lower end of the tube D, and secured thereto by a pin, *o'*, or like device. This knob is bored to admit the passage of the push-rod F through it, and has a central recess to receive a spring, *p*, which is spirally coiled around the push-rod and bears against the shank of the button *f*, which is screwed onto the end of the push-rod.

The operation is as follows: To disengage the cross-piece *g* and rack E, press upon the button *f*, which in turn forces the wedge-shaped end of the rod F between the smooth edge of the rack and the rounded cross-piece *g'*, thereby moving the parts G G' and lifting the cross-piece *g* clear of the teeth *b*. The tube D may then be raised or lowered, as may be desired. When the pressure is taken off the button *f* the rod F is carried downwardly and out of contact with the cross-piece *g'* by the action of the spring *p* on the button *f*, and the cross-piece *g* enters one of the indentations between the teeth *c* under an impulse from the spring *j*, and the chandelier is locked and held stationary at the desired position. The downward pull to extend the tube D unwinds the suspending-cord *b* from the drum C, and when the tube is pushed upwardly the spring within the drum C causes the cord to be rewound upon said drum, and thus a proper degree of tension on the cord is always secured.

It will be seen that the extension of the tube D must stop when the buffer *e* comes in contact with the catch G G', as the buffer *e* is too large to pass through the opening between the parts G G'.

What I claim as my invention, and desire to secure by Letters Patent, is—

1. The combination of a stationary or non-extensible tube depending from a pipe or nipple fixed to the yoke carrying the spring-drum,

a stationary rack or indented bar centrally fixed in said stationary tube, an extensible tube fitted within said stationary tube and adapted to slide therein, and a suspending-cord, one end of which is fixed to the spring-drum and the other to the extensible tube, with a catch or locking device pivoted within said extensible tube, and adapted to be disengaged from the rack by said rod and engaged with said rack by a spring, substantially as herein shown and described.

2. The combination of a stationary or non-extensible tube, a stationary rack or indented bar, an extensible tube, suspending-cord, and vertical rod, with a locking device pivoted to the prong *h* of an annular ring, H, fixed to and adapted to move with the tube D, and bearing against a spring, *j*, whereby the extensible tube is locked by the action of said spring and released by the action of said vertical rod, substantially as herein shown and described.

3. The combination, in a chandelier having a stationary or non-extensible tube, a stationary rack or indented bar, an extensible tube, suspending-cord, and spring-drum, of the wedge-shaped vertical rod F, with the catch or locking device G G', pivoted to the prong *h*, and provided with cross-pieces *g g'*, and the spring *j*, fixed to the prong *h'*, and bearing against a stud, *k*, whereby the piece *g* is engaged with the rack to lock the extensible tube by the action of the spring, and is disengaged from the rack to permit the movement of the extensible tube by the upward thrust of the rod, substantially as herein shown and described.

4. The combination, in a chandelier having a stationary or non-extensible tube, a stationary rack or indented bar, an extensible tube, suspending-cord, and spring-drum, of the rod F, recessed knob *o*, spring *p*, and button *f*, with the catch or locking device G G', pivoted to the prong *a*, spring *j*, fixed to the prong *h'*, and stud *k*, substantially as herein shown and described, for the purposes set forth.

In testimony whereof I have hereunto set my hand this 6th day of March, A. D. 1883.

JOHN T. BRUEN.

Witnesses:

ARTHUR C. WEBB,
HERBERT SOUTHRUCK.